Mid-term Project for Optimization Course

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April 4, 2026

For the mid-term project, you are asked to work on one of the models in this reading list note. The reading list here collects *static* fertility choice problems from research papers and from Hannusch and Yum (2026)[1] and Doepke et al (2023)[2] .

Each section is written so that you can understand **why** the model exists, **what** economic relationship it focuses on, and **how** the mathematics encodes that story. I include the original papers in the footnotes from which the model was taken. While it is not required to read them to follow the logic here, you are encouraged to do so.

Your task:

1. Pick one model of your choice
2. Solve the model and derive the main results
3. Read the full paper or relevant excerpt or book chapter to understand the model and its application in the paper. Collect data to demonstrate the author's point.
4. Prepare a short presentation slide, either in Vietnamese or English
   - The slides must be no longer than 10 pages
   - Font size must be 13pt or above
   - Each presentation is at most 10 minutes, with Q&A of 5 minutes.
5. Required content
   - Model's main assumption and results
   - Present some data, empirical evidence, or articles that can highlight the model's insights

If you have any questions, send an email to me here:

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[1] Anne Hannusch and Minchul Yum (2026), "Economic Theories of Fertility Choice," chapter draft (March 2026), prepared for *The Oxford Handbook of Fertility and Reproductive Health Economics*, edited by Kelly M. Jones and Damian Clarke.

[2] Doepke, M., Hannusch, A., Kindermann, F., & Tertilt, M. (2023). The economics of fertility: A new era. In Handbook of the Economics of the Family (Vol. 1, No. 1, pp. 151-254). North-Holland.

1 Becker (1960): subsistence consumption and fertility

Motivation. In rich countries, richer households often have *fewer* children than poorer households. A naive model with Cobb–Douglas preferences over consumption and children and a single time constraint makes fertility *independent* of the wage, because income and substitution effects from a wage increase cancel under log utility. So the model fails the most basic fact.

Mechanism. Becker's minimal fix is **subsistence consumption** $\bar{c}$: utility depends on $\ln(c+\bar{c})$ instead of $\ln c$. When the wage is low, much of market income is needed just to push consumption above subsistence; time spent on children is "cheap" in forgone consumption. When the wage rises, the opportunity cost of time in childrearing rises in *relative* terms, so parents shift time toward the market and have fewer children. The model is deliberately static: one period, one choice of how to split time between market work and childrearing.

References. Hannusch and Yum (2026, Sec. 2.1) use this as the textbook benchmark; see also Jones, Schoonbroodt, and Tertilt (2010) on mechanisms for the income–fertility gradient.[3]

Variables and parameters. c: consumption; $n > 0$: children; $\ell \in [0, 1]$: time in childrearing; $w > 0$: wage; $\bar{c} > 0$: subsistence (minimum psychological or biological need for consumption above which utility is "normal"); $\phi > 0$: productivity of time in childrearing (births per unit ℓ); $\theta \in (0, 1)$: taste weight on consumption versus children.

Objective.

$$\max_{c,\, n} \quad \theta \ln(c + \bar{c}) + (1 - \theta) \ln n.$$

Constraints.

- $c = (1 - \ell)w$ (market time $1 - \ell$).
- $n = \phi\ell$.
- $0 \leq \ell \leq 1$.

1. Show that the interior first-order condition implies

$$n^* = \phi(1 - \theta)\Big(1 + \frac{\bar{c}}{w}\Big).$$

2. Show that $\dfrac{\partial n^*}{\partial w} = -\phi(1 - \theta)\dfrac{\bar{c}}{w^2} < 0$ and interpret.

[3] Gary S. Becker (1960), "An economic analysis of fertility," in *Demographic and Economic Change in Developed Countries*, NBER, 209–240. Robert W. Jones, Alice Schoonbroodt, and Michèle Tertilt (2010), "Fertility theories: can they explain the negative fertility-income relationship?" *Journal of Demographic Economics* 76(1): 28–55.

2 Child quality and quantity (De la Croix–Doepke)

Motivation. The negative correlation between income and fertility coexists with the fact that richer parents spend *more per child* on education and health. Becker and Lewis (1973) framed this as a **quantity–quality trade-off**: children are costly both in time and in goods; raising "quality" (human capital) per child competes with having more children for the same budget.

Mechanism. Parents care about $\ln(nh)$—a stand-in for utility from family size weighted by child quality—while quality h is produced by spending e on each child. Higher wages relax the budget but also increase the opportunity cost of time ϕn. In interior solutions, a higher wage can increase optimal e (quality) and reduce optimal n (quantity). De la Croix and Doepke (2003) show how differential fertility across rich and poor families feeds back into inequality and growth; here we only need the static household block.

References.[4]

Variables and parameters. c: parental consumption; $n > 0$: number of children (fertility); $e \geq 0$: education input per child; h: child "quality" (human capital); w: wage (market time endowment normalized); p: price of education; ϕ: time needed per child; $\theta \geq 0$: baseline human capital without parental spending; $\gamma \in (0, 1)$: elasticity of quality in $e + \theta$; $\delta > 0$: taste for children. Parameters $w, p, \phi, \theta, \delta, \gamma$ as above.

Objective.

$$\max_{c,\, n,\, e} \quad \ln c + \delta \ln(nh), \qquad h = (e + \theta)^{\gamma}.$$

Constraints.

- Budget: $c + pen = (1 - \phi n)w$.
- Nonnegativity: $c > 0$, $n > 0$, $e \geq 0$.
- Feasibility of fertility: $\phi n < 1$ (mother has enough time).

1. Show that optimal consumption is always $c^* = w/(1 + \delta)$. Show that if $w > \theta p/(\gamma\phi)$ then

$$e^* = \frac{\gamma\phi w/p - \theta}{1 - \gamma}, \qquad n^* = \frac{\delta}{1 + \delta} \frac{1 - \gamma}{\phi w - \gamma p \theta},$$

whereas if $w \leq \theta p/(\gamma\phi)$ then $e^* = 0$ and $n^* = \dfrac{\delta}{1 + \delta} \dfrac{1}{\phi}$.

2. Assume the interior regime $w > \theta p/(\gamma\phi)$. Show that

$$\frac{\partial n^*}{\partial w} < 0, \qquad \frac{\partial n^*}{\partial \phi} < 0.$$

Briefly interpret why a larger time cost ϕ lowers fertility.

[4] David de la Croix and Matthias Doepke (2003), "Inequality and growth: why differential fertility matters," *American Economic Review* 93(4): 1091–1113. Gary S. Becker and Gregg H. Lewis (1973), "On the interaction between the quantity and quality of children," *Journal of Political Economy* 81(2, Part 2): S279–S288.

3 Separable quantity–quality

Motivation. The previous section uses a multiplicative form $h = (e + \theta)^\gamma$. Many textbooks and policy discussions use a **linear** quality technology: spending s per child adds to quality $q = a + \delta s$. Separable logarithmic utility $\theta_c \ln c + \theta_n \ln n + \theta_q \ln q$ then delivers clean closed forms for investment s and fertility n, which is convenient for comparative statics and calibration.

Mechanism. The same tension as Becker–Lewis appears: resources spent on s per child (times n) trade off against consumption and against time devoted to child quantity. The condition $\theta_n > \theta_q$ ensures that the trade-off between quantity and quality is well-behaved (quality does not "dominate" quantity in a way that would overturn the problem).

References. Hannusch and Yum (2026, Sec. 2.2); Becker and Tomes (1976); Willis (1973).[5]

Variables and parameters. c, s, n, q, ℓ as below; $\theta_c, \theta_n, \theta_q > 0$ with $\theta_c + \theta_n + \theta_q = 1$; $a \geq 0$, $\delta > 0$; $w, \phi > 0$.

Objective.

$$\max_{c,\,n,\,s,\,q} \quad \theta_c \ln c + \theta_n \ln n + \theta_q \ln q.$$

Constraints.

- $c + sn = (1 - \ell)w$.
- $n = \phi\ell$.
- $q = a + \delta s$.
- $0 \leq \ell \leq 1$.

1. Under $\theta_n > \theta_q$ and $\delta w > a\phi$, show that interior first-order conditions imply the handbook formulas

$$s^* = \frac{1}{\theta_n - \theta_q}\Big(\frac{\theta_q w}{\phi} - \frac{\theta_n a}{\delta}\Big), \qquad n^* = \frac{\theta_n - \theta_q}{1 - \theta_q}\,\frac{\delta\phi}{\delta w - a\phi}\,w.$$

2. Show that $\partial n^*/\partial w < 0$ and $\partial s^*/\partial w > 0$ (canonical quantity–quality trade-off).

[5] Gary S. Becker and Nigel Tomes (1976), "Child endowments and the quantity and quality of children," *Journal of Political Economy* 84(4, Part 2): S143–S162. Robert J. Willis (1973), "A new approach to the economic theory of fertility behavior," *Journal of Political Economy* 81(2, Part 2): S14–S64.

4 Longevity, saving, and fertility

Motivation. The demographic transition was accompanied by rising life expectancy. If people expect to live longer, they may value retirement consumption more and allocate more resources to old age. That can crowd out current consumption and childbearing in a static budget sense, or shift marginal rates of substitution.

Mechanism. This two-period stylization puts old-age consumption c_{old} in the utility with weight π (probability of surviving to enjoy it). The budget $c + \pi c_{\text{old}} = (1 - \ell)w$ says that delivering one unit of consumption when old costs π in present value when young (or equivalently, only a fraction π of savings pay off). Higher π raises the attractiveness of shifting resources toward c_{old}, reducing time ℓ for children and thus fertility $n = \phi\ell$. The key point: **with self-financed retirement**, longevity and fertility move in opposite directions in this benchmark.

References. Hannusch and Yum (2026, Sec. 2.3); Zhang, Zhang, and Lee (2003); Cervellati and Sunde (2005); Soares (2005).[6]

Variables and parameters. c, c_{old}: consumption when young and old; n: children; ℓ; $\pi \in [0, 1]$: survival to old age; $\theta \in (0, 1)$; w, ϕ as usual.

Objective.

$$\max_{c, c_{old}, \ell} \quad \theta\big(\ln c + \pi \ln c_{\text{old}}\big) + (1 - \theta)\ln n.$$

Constraints.

- $c + \pi c_{\text{old}} = (1 - \ell)w$.
- $n = \phi\ell$.
- $0 \leq \ell \leq 1$.

1. Show that optimal fertility satisfies

$$n^* = \phi\,\frac{1 - \theta}{1 + \pi\theta},$$

and hence $\partial n^*/\partial\pi < 0$.

[6] Junsen Zhang, Jie Zhang, and Ronald Lee (2003), "Rising longevity, education, savings, and growth," *Journal of Development Economics* 70(1): 83–101. Matteo Cervellati and Uwe Sunde (2005), "Human capital formation, life expectancy, and the process of development," *American Economic Review* 95(5): 1653–1672. Rodrigo R. Soares (2005), "Mortality reductions, educational attainment, and fertility choice," *American Economic Review* 95(3): 580–601.

5 Children as old-age insurance

Motivation. The previous section assumes parents save for old age. Historically (and in many developing economies), formal saving and public pensions are limited; **children** are an informal insurance asset—transfers v per child in old age. Then the comparative statics of longevity on fertility can *flip*.

Mechanism. If $c_{\text{old}} = vn$, more children directly raise old-age consumption. A higher survival probability π increases the expected utility from those transfers, so the value of having children rises. Introducing a public pension ψ that adds to c_{old} without requiring more children weakens the private return to fertility and tends to reduce n. This logic underlies arguments that social security crowds out the "demand for children" as old-age support.

References. Boldrin and Jones (2002); Ehrlich and Kim (2007).[7]

Objective.

$$\max_{c,\ell} \quad \theta\big(\ln c + \pi \ln c_{\text{old}}\big) + (1-\theta)\ln n.$$

Constraints.

- $c = (1-\ell)w, \quad c_{\text{old}} = vn$ with $v > 0$.
- $n = \phi\ell, \quad 0 \le \ell \le 1$.

1. Show that

$$n^* = \phi\,\frac{1-\theta+\theta\pi}{1+\theta\pi},$$

so $\partial n^*/\partial\pi > 0$.

2. (Extension) With $c_{\text{old}} = vn + \psi$, $\psi \ge 0$, where ψ is the public pension. Show that $\mathrm{d}n^*/\mathrm{d}\psi < 0$ at an interior solution.

[7] Michele Boldrin and Larry E. Jones (2002), "Mortality, fertility, and saving in a Malthusian economy," *Review of Economic Dynamics* 5(4): 775–814. Isaac Ehrlich and Jinyoung Kim (2007), "Social security and demographic trends: theory and international evidence," *Review of Economic Dynamics* 10(1): 55–77.

6 Child mortality and births

Motivation. Demographers long argued that as child mortality falls, parents need fewer births to reach a *target* number of surviving children. Economists asked whether this channel alone can explain the large fertility declines observed in the demographic transition.

Mechanism. Utility depends on $\ln(sn)$: parents care about *expected surviving* children $S \equiv sn$. Time ℓ produces $S = \phi\ell$ (a reduced form). Better survival s means fewer births $n = S/s$ are needed to achieve the same S. In this stripped-down static model, **expected surviving fertility** S^* is pinned down by preferences and wages alone, so gross births n^* fall one-for-one with s when S^* is fixed—but S^* itself is independent of s. Thus mortality decline alone cannot explain a large drop in S; other mechanisms (quality–quantity, opportunity costs, etc.) are needed for the full story (Doepke, 2005).

References. Hannusch and Yum (2026, Sec. 2.4); Eckstein, Wolpin, and Mira (1999); Doepke (2005).[8]

Variables and parameters. $s \in (0, 1)$: exogenous survival probability per birth; n: births; expected surviving children sn.

Objective. (Hannusch and Yum, 2026, Sec. 2.4)

$$\max_{c,n,\ell} \quad \theta \ln c + (1-\theta)\ln(sn).$$

Constraints.

- $c = (1-\ell)w$.
- $sn = \phi\ell$ (time ℓ yields expected surviving children $\phi\ell$).
- $0 \leq \ell \leq 1$.

1. Let $S \equiv sn = \phi\ell$. Show that the problem reduces to choosing $S \in (0, \phi]$ and yields $S^* = (1-\theta)\phi$, hence births $n^* = \phi(1-\theta)/s$. Show $\partial n^*/\partial s < 0$ but sn^* does not depend on s.

[8] Zvi Eckstein, Kenneth I. Wolpin, and Pedro R. Mira (1999), "A dynamic analysis of schooling, fertility, and child mortality," *Journal of Political Economy* 107(4): 786–820. Matthias Doepke (2005), "Child mortality and fertility decline: Does the Barro–Becker model fit the facts?" *Journal of Population Economics* 18(2): 337–366.

7 Gender wage gap and fertility

Motivation. Women's wages have risen relative to men's over the long run in many countries. If mothers provide a fixed share ϕ of childcare time per child, a higher female wage raises the **dollar opportunity cost** of each child more than it raises family income (because the husband's wage may not move in the same proportion).

Mechanism. The budget $c + \phi w_f n = w_m + w_f$ makes the effective price of a child proportional to w_f. A rise in w_f increases full income $w_m + w_f$ (income effect toward more of everything, including children) but increases the marginal cost ϕw_f even faster when the wife's time is the margin (substitution effect toward fewer children). In this log-utility, linear-budget specialization, the substitution effect wins for n: fertility falls when the wife's wage rises holding w_m fixed, and rises when the husband's wage rises. Narrowing the gender gap (raising w_f toward w_m) therefore pushes *down* fertility in this benchmark—unless other margins (market childcare, father's time) relax.

References. Galor and Weil (1996); Doepke *et al.* (2023, Sec. 2.4).[9]

Variables and parameters. c: couple's consumption; $n > 0$: fertility; w_m, w_f: husband's and wife's wages; ϕ: time per child borne by the wife; $\delta > 0$: taste for children. (The full model allows child quality; here $\gamma = 0$ in the quality term so only quantity enters utility.)

Objective.

$$\max_{c,\,n} \quad \ln c + \delta \ln n.$$

Constraints.

- Budget: $c + \phi w_f\, n = w_m + w_f$.
- Positivity: $c > 0$, $n > 0$.

1. Show that

$$c^* = \frac{w_m + w_f}{1+\delta}, \qquad n^* = \frac{\delta}{1+\delta}\,\frac{1}{\phi}\left(1 + \frac{w_m}{w_f}\right).$$

2. Show that $\partial n^*/\partial w_f < 0$ and $\partial n^*/\partial w_m > 0$. If w_f rises toward w_m (narrowing the gender gap) holding w_m fixed, does fertility rise or fall?

[9] Oded Galor and David N. Weil (1996), "The gender gap, fertility, and growth," *American Economic Review* 86(3): 374–387. Matthias Doepke *et al.* (2023, Sec. 2.4).

8 Marketization of childcare

Motivation. In the 1980s, female employment and fertility were negatively correlated across OECD countries; in the 2000s, some high–female-employment countries also had relatively high fertility. One hypothesis is that **cheap, available market childcare** (or fathers' time) uncouples fertility from the mother's wage.

Mechanism. Let s be the fraction of required childcare time bought at money price p_s per unit instead of supplied by the mother at opportunity cost w_f. The effective marginal cost of children becomes a blend of w_f and p_s. When s is high enough, a higher w_f can raise fertility because income rises while the mother's time is less binding (the derivative $\partial n^*/\partial w_f$ can turn positive). The threshold ω in the exercises is the point where the "income" and "opportunity cost" forces balance.

References. Ahn and Mira (2002); Hazan and Zoabi (2015); Doepke *et al.* (2023, Sec. 4.2).[10]

Variables and parameters. c, n: consumption and fertility; $s \in [0, \bar{s}]$: share of childcare bought on the market; p_s: monetary price per unit of outsourced childcare; ψ: goods cost per child; ϕ: total childcare time per child; w_m, w_f: wages; $\delta > 0$: taste for children.

Objective.

$$\max_{c,\,n} \quad \ln c + \delta \ln n.$$

Constraints. (for a fixed policy or equilibrium value of s)

- Budget:
$$c + \psi n + s p_s n \phi = w_m + w_f \big[1 - (1-s) n \phi\big].$$
- Positivity: $c > 0$, $n > 0$.

1. Show that

$$c^* = \frac{w_m + w_f}{1+\delta}, \qquad n^*(s) = \frac{\delta}{1+\delta} \, \frac{w_m + w_f}{\psi + \big[s p_s + (1-s) w_f\big]\phi}.$$

2. Show that

$$\frac{\partial n^*}{\partial w_f} = \frac{\delta}{1+\delta} \, \frac{\psi + \big(s p_s - (1-s) w_m\big)\phi}{\big(\psi + [s p_s + (1-s) w_f]\phi\big)^2}.$$

Assume $\psi < w_m \phi$. Show that $\partial n^*/\partial w_f < 0$ if and only if $s < \omega$, that $\partial n^*/\partial w_f = 0$ if $s = \omega$, and that $\partial n^*/\partial w_f > 0$ if $s > \omega$, where

$$\omega = \frac{w_m \phi - \psi}{(p_s + w_m)\phi}.$$

[10] Namkee Ahn and Pedro Mira (2002), "A note on the changing relationship between fertility and female employment rates in developed countries," *Journal of Population Economics* 15(4): 667–682. Moshe Hazan and Hosny Zoabi (2015), "Do highly educated women choose smaller families?" *Economic Journal* 125(587): 1191–1226. Matthias Doepke *et al.* (2023, Sec. 4.2).

9 Easterlin: relative income and the baby boom

Motivation. The U.S. baby boom (mid-20th century) is hard to explain with a pure income story: incomes were rising, yet fertility rose. Easterlin proposed that fertility depends on **relative** income—wages relative to the consumption standards people absorbed from their parents' generation.

Mechanism. Utility from consumption depends on $c - \chi c_{-1}$: only consumption *above* a habit or aspiration level (linked to parental consumption c_{-1}) yields utility. When current wages w are high compared to the reference implied by c_{-1}, households feel "prosperous" relative to their aspirations and allocate more time to childrearing (higher n). When the same generation later faces cohort crowding and lower wages relative to aspirations, fertility can fall—a story of the boom and bust tied to relative economic position, not the level of income alone.

References. Easterlin (1961); habit formation in Abel (1990); Hannusch and Yum (2026, Sec. 2.6).[11]

Variables and parameters. $c_{-1} > 0$: reference consumption (parental background); $\chi \in (0, 1)$: weight on aspirations.

Objective.

$$\max \quad \theta \ln(c - \chi c_{-1}) + (1 - \theta) \ln n.$$

Constraints.

- $c = (1 - \ell)w, \quad n = \phi\ell, \quad 0 \le \ell \le 1.$
- Interiority: $c > \chi c_{-1}$.

1. Show that

$$n^* = (1-\theta)\,\phi \left[1 - \chi\Big(\frac{w}{c_{-1}}\Big)^{-1}\right] = (1-\theta)\,\phi\,\Big(1 - \frac{\chi c_{-1}}{w}\Big).$$

2. Show that $\partial n^*/\partial w > 0$ holding c_{-1} fixed (a baby boom when young cohorts enjoy wages high relative to aspirations).

[11] Richard A. Easterlin (1961), "The American baby boom in historical perspective," *American Economic Review* 51(5): 869–898. Andrew B. Abel (1990), "Asset prices under habit formation and catching up with the Joneses," *American Economic Review* 80(2): 38–42.

10 Fehr–Ujhelyiova: static model of fertility choice

Motivation. Fehr and Ujhelyiova (2013) build a large quantitative model of German families to study taxes, benefits, and childcare policy. Before launching that machinery, they need a transparent static household core: how do money costs b_0 and time costs b_1 of children interact with wages and preferences?

Mechanism. The budget $c+b_0n = w(1-b_1n)$ says full income w is split between consumption and children, where the "full price" of a child is $b_0 + wb_1$ (goods plus forgone market time). CES preferences between c and n allow substitution elasticity ρ to govern whether income raises or lowers n when only time costs matter ($b_0 = 0$), as Fehr and Ujhelyiova discuss on their page 4. Policy can lower b_0 (transfers) or b_1 (subsidized childcare).

References. Fehr and Ujhelyiova (2013, Sec. 2 and p. 4).[12]

Variables and parameters. $c > 0$: consumption; $n > 0$: number of children; $a_c, a_n > 0$: taste weights; $\rho > 0$, $\rho \neq 1$: substitution elasticity; $w > 0$: wage; $b_0 \geq 0$, $b_1 \geq 0$: money and time costs per child.

Objective. (Fehr and Ujhelyiova 2013, p. 4)

$$\max_{c,\,n} \quad U(c,n) = a_c\,\frac{c^{1-1/\rho}}{1-1/\rho} \;+\; a_n\,\frac{n^{1-1/\rho}}{1-1/\rho}.$$

Constraints.

- $c + b_0n = w(1 - b_1n) \quad \Leftrightarrow \quad c + (b_0 + wb_1)\,n = w.$
- Interiority: $c > 0$, $n > 0$, $1 - b_1n > 0$.

1. With multiplier λ on $w - c - (b_0 + wb_1)n$, show $a_cc^{-1/\rho} = \lambda$ and $a_nn^{-1/\rho} = \lambda(b_0 + wb_1)$.
2. Show

$$n^* \;=\; \frac{w}{\left(\dfrac{a_c}{a_n}(b_0 + wb_1)\right)^{\rho} \;+\; b_0 + wb_1}\;.$$

3. For $p \equiv b_0 + wb_1$, show $\partial n^*/\partial b_0 < 0$ and $\partial n^*/\partial b_1 < 0$ at an interior optimum (fixed w).

[12] Hans Fehr and Daniela Ujhelyiova (2013), "Fertility, female labor supply, and family policy," *German Economic Review* 14(2): 138–165, `doi:10.1111/j.1468-0475.2012.00568.x`.

11 Workplace inflexibility and fertility

Motivation. High-paying jobs often require long hours, fixed schedules, or face time. Combining those jobs with young children is especially costly. Goldin (2014) documents how "greedy jobs" and inflexible hours interact with the gender division of childcare.

Mechanism. The term ηwn in the time constraint is an extra time burden per child that scales with the wage (or with job demands): inflexibility raises the effective hours lost per child beyond basic childcare ℓ. Higher η or higher w tightens the feasibility set and lowers equilibrium fertility n^* in the closed form below. The model is a reduced form for a rich policy debate: flexible schedules, parental leave, and part-time entitlements all aim to lower this implicit η.

References. Hannusch and Yum (2026, Sec. 3.3); Goldin (2014); Guner, Ventura, and Okou (2024).[13]

Variables and parameters. h, ℓ: market work and home time devoted to children; $\eta > 0$: inflexibility shifter; $\gamma, \theta > 0$ with $\gamma + \theta < 1$.

Objective.

$$\max_{c,n,\ell,h} \quad \theta \ln c + \gamma \ln n + (1 - \theta - \gamma) \ln\bigl(1 - h - \ell - \eta wn\bigr).$$

Constraints.

- $c = wh$.
- $n = \phi\ell$.
- $h, \ell \geq 0, \quad h + \ell + \eta wn \leq 1$.

1. Show that interior optimization yields

$$n^* = \frac{\gamma}{\frac{1}{\phi} + \eta w},$$

and hence $\partial n^*/\partial \eta < 0$ and $\partial n^*/\partial w < 0$.

[13] Claudia Goldin (2014), "A grand gender convergence: its last chapter," *American Economic Review* 104(4): 1091–1119. Nezih Guner, Gustavo Ventura, and Yamèle Okou (2024), "Labor market institutions and fertility," working paper.

12 Careers and timing of fertility

Motivation. Mean age at first birth has risen in most rich countries. Delaying fertility can protect early-career human capital $h(e)$, but biological fecundity declines with age, so late attempts may fail with probability $1-\pi$. Women therefore face a trade-off between "early" and "late" childbearing.

Mechanism. This stylized two-period block fixes discrete fertility plans $(n_1, n_2) \in \{0,1\}^2$ and solves for optimal career effort e when young. Early birth reduces time available for e; late birth risks not receiving the child at all. Comparing indirect utilities $u_{1,0}$, $u_{0,1}$, etc., spells out when delay is optimal. The full dynamic models in Caucutt, Guner, and Knowles (2002) add marriage markets and wage processes; here we only need the static trade-off embedded in e and π.

References. Doepke *et al.* (2023, Sec. 4.3); Caucutt, Guner, and Knowles (2002); Goldin (2014).[14]

Variables and parameters. $e \in [0,1]$: time invested in career when young; $n_1, n_2 \in \{0,1\}$: child in period 1 or 2 (at most one child total); $c_1, c_{2,0}, c_{2,1}$: consumption when young, old without late child, old with late child; w_f: wage; κ, γ: scale and elasticity of second-period human capital $h(e) = \kappa e^\gamma$; ϕ: time cost of a child; $\pi \in (0,1)$: probability late birth succeeds; ν: utility from a child.

Objective. Given (n_1, n_2), maximize expected utility by choosing e:

$$\max_e \ \ln(c_1) + (1-\pi n_2)\ln(c_{2,0}) + \pi n_2 \ln(c_{2,1}) + \nu n_1 + \pi \nu n_2.$$

Constraints.

- $c_1 = w_f(1 - n_1\phi - e)$.
- If $n_2 = 0$: second-period consumption is $c_{2,0} = w_f \kappa e^\gamma$.
- If $n_2 = 1$: $c_{2,1} = w_f \kappa e^\gamma (1-\phi)$ when birth succeeds (and $c_{2,0}$ applies with probability $1-\pi$ as in the objective).

1. Show that, given n_1, n_2, the optimal career choice is $e = \frac{\gamma}{1+\gamma}(1 - n_1\phi)$.
2. Take $n_1 = n_2 = 0$, show the optimal utility is $u_{0,0} = \ln\big(\frac{w_f}{1+\gamma}\big) + \ln\big(\kappa w_f \big(\frac{\gamma}{1+\gamma}\big)^\gamma\big)$.
3. Take $n_1 = 1, n_2 = 0$, show $u_{1,0} = u_{0,0} + (1+\gamma)\ln(1-\phi) + \nu$.
4. Take $n_1 = 0, n_2 = 1$, show $u_{0,1} = u_{0,0} + \pi[\ln(1-\phi) + \nu]$.
5. Under what condition on ν does having children early rather than late yield higher utility?

[14] Elizabeth M. Caucutt, Nezih Guner, and John Knowles (2002), "Why do women wait? Matching, wage inequality, and the incentives for fertility delay," *Review of Economic Dynamics* 5(4): 815–855. Claudia Goldin (2014), *American Economic Review* 104(4): 1091–1119.

13 Childlessness

Motivation. Completed fertility in rich countries includes a large share of women with *zero* children. Empirically, childlessness is sometimes U-shaped in education: high among the poorest (cannot afford a fixed cost) and among the richest (very high opportunity cost). Continuous fertility models miss the **extensive margin** $n = 0$ vs. $n > 0$.

Mechanism. Baudin *et al.* (2015) combine three devices: (i) a fixed consumption floor c_f required to "activate" motherhood (goods and services); (ii) a wage penalty $\tau w_f^2 n$ that makes combining children and careers especially costly when female wages are high (Goldin, 2014); (iii) a small $\nu > 0$ so that $\ln(n+\nu)$ is well-defined at $n = 0$ and utility is continuous. Compare utility U_0 at $n = 0$ with U_{int} at an interior n to obtain a participation condition.

References. Hannusch and Yum (2026, Sec. 3.4); Baudin, Docquier, and Moizeau (2015); Baudin, de la Croix, and Gobbi (2015); Goldin (2014).[15]

Variables and parameters. c: consumption; $n \geq 0$: fertility; $\ell \in [0,1]$: mother's time in childrearing; $w_m, w_f > 0$: male and female wages; $\phi > 0$: fertility per unit ℓ; $c_f > 0$: minimum consumption if $n > 0$; $\nu > 0$: utility shifter; $\tau > 0$: strength of the career penalty; $\theta \in (0,1)$.

Objective. If $n > 0$, utility is U_{int}; if $n = 0$, utility is U_0 (below).

$$U_{\text{int}} = \theta \ln c + (1-\theta)\ln(n+\nu).$$

Constraints.

- Budget with indicator $\mathbb{I}(n > 0)$ (mother works $1-\ell$; father earns w_m):

$$c + c_f\,\mathbb{I}(n > 0) = (1-\ell)w_f + w_m - \tau w_f^2\, n.$$

- $n = \phi\ell, \quad 0 \leq \ell \leq 1.$
- If $n = 0$: $\ell = 0$ and $c = w_m + w_f$.

1. With $n = \phi\ell > 0$, substitute $\ell = n/\phi$ and show that

$$c = w_m + w_f - c_f - n\Big(\frac{w_f}{\phi} + \tau w_f^2\Big).$$

2. Show that an interior choice $n > 0$ satisfies the first-order condition (derive $\partial U_{\text{int}}/\partial n = 0$) and the closed form

$$n^*_{\text{int}} = \frac{(1-\theta)(w_m + w_f - c_f)}{\dfrac{w_f}{\phi} + \tau w_f^2} - \theta\nu.$$

3. With $n = 0$, $U_0 = \theta \ln(w_m + w_f) + (1-\theta)\ln\nu$. Show that the extensive margin is

$$n^* = \begin{cases} 0, & \text{if } U_0 > U_{\text{int}}(n^*_{\text{int}}) \text{ (or } n^*_{\text{int}} \leq 0), \\ n^*_{\text{int}}, & \text{otherwise.} \end{cases}$$

4. (Comparative statics.) Show $\partial n^*_{\text{int}}/\partial w_f$ has ambiguous sign in general (opportunity cost vs. income); show $\partial n^*_{\text{int}}/\partial\tau < 0$ when $n^*_{\text{int}} > 0$.

[15] Thomas Baudin, David Docquier, and Fabien Moizeau (2015), "Fertility choices and sectoral reallocation: theory and evidence," working paper; see also Thomas Baudin, David de la Croix, and Paula E. Gobbi (2015), "Fertility and childlessness in the US," *American Economic Review: Papers & Proceedings* 105(5): 533–538. Claudia Goldin (2014), "A grand gender convergence: its last chapter," *American Economic Review* 104(4): 1091–1119.

14 Customary law: Polygyny

Motivation. In parts of Sub-Saharan Africa and elsewhere, customary law allows **polygyny**: one husband, multiple wives. Institutions also set **bride prices** paid to the wife's family. Tertilt (2005) argues these features interact with fertility and savings in general equilibrium; here we use a static "toy" that isolates one trade-off.

Mechanism. Total children n are costly in a **convex** way when borne within one marriage: the resource cost is modeled as n^2/f, which falls when more wives f share the burden—more wives spread children and reduce the marginal congestion cost per birth. That benefit must be weighed against the linear cost pf of bride prices. When p is low relative to income y, optimal f^* exceeds one (polygyny). The functional form is a reduced form for sibling competition and household resources; the point is the economic tension, not literal biology.

References. Hannusch and Yum (2026, Sec. 4.1); Tertilt (2005).[16]

Variables and parameters. $y > 0$: male income; c: consumption; $n > 0$: total children; $f > 0$: number of wives; $p > 0$: bride price per wife; $\theta \in (0,1)$.

Objective.

$$\max_{c,\, n,\, f} \quad \theta \ln c + (1-\theta) \ln n.$$

Constraints.

- Budget: $c + pf + \dfrac{n^2}{f} = y$.
- $c > 0$, $n > 0$, $f > 0$ (interior).

1. Show that first-order conditions imply

$$f^* = \frac{(1-\theta)\, y}{2p}, \qquad n^* = \frac{(1-\theta)\, y}{2\sqrt{pf^*}}.$$

2. Conclude that $n^* = \sqrt{\dfrac{(1-\theta)\, y}{2}}$ and that **polygyny** ($f^* > 1$) occurs if and only if $p < \dfrac{(1-\theta)\, y}{2}$.

3. Show that $\partial f^*/\partial p < 0$ and $\partial n^*/\partial y > 0$.

[16] Michèle Tertilt (2005), "Polygyny, fertility, and savings," *Journal of Political Economy* 113(6): 1381–1411. Paula Gobbi, David de la Croix, and Michèle Tertilt (2026), "Fertility in developing countries," in *Oxford Handbook of Fertility and Reproductive Health Economics* (forthcoming).

15 Impartible vs. partible inheritance

Motivation. Land was the main asset in pre-industrial agriculture. **Partible** inheritance splits the plot among all children; **impartible** inheritance assigns the whole plot to one heir (often the eldest son). Historical evidence links these rules to fertility: smaller plots under partible division can reduce per-heir productivity when land has increasing returns to scale at the plot level.

Mechanism. Parents care about $\ln(ny'_i)$ where aggregate children's income ny'_i equals $A(L)n$ under impartible rules (joint work on the undivided farm) but $A(L)n^{\alpha}$ with $\alpha < 1$ under partible rules (fragmentation destroys output more than proportionally). Lower α makes extra children dilute land per heir more sharply, reducing the marginal benefit of n and lowering optimal fertility relative to impartible inheritance. The model rationalizes why ethnic or regional inheritance customs can shift aggregate fertility even holding wages fixed.

References. Hannusch and Yum (2026, Sec. 4.1); Fontenay, Gobbi, and Sangnier (2025); Gay, Gobbi, and Sangnier (2025).[17]

Variables and parameters. $y > 0$: parental income; $L > 0$: land endowment; $A(L) > 0$: land productivity shifter; $\alpha \in (0, 1)$: elasticity of fragmentation loss under partible rules; $\phi > 0$; $\theta \in (0, 1)$.

Objective. In each regime $i \in \{I, P\}$,

$$\max_{c,\, n,\, \ell} \quad \theta \ln c + (1-\theta) \ln\big(ny'_i\big),$$

where children's aggregate future income ny'_i depends on the inheritance institution. **Constraints.**

- $c = (1-\ell)y, \quad n = \phi\ell, \quad 0 \le \ell \le 1.$
- **Impartible (I):** $ny'_I = A(L)\, n$ (no fragmentation; children work jointly).
- **Partible (P):** $ny'_P = A(L)\, n^{\alpha}$ (output loss from dividing the plot).

1. Under impartible inheritance, show that

$$n_I^* = (1-\theta)\,\phi.$$

2. Under partible inheritance, show that

$$n_P^* = (1-\theta)\,\phi\,\frac{\alpha}{\theta + (1-\theta)\alpha},$$

and verify that $n_P^* < n_I^*$ when $\alpha < 1$.

3. Interpret: why does partible inheritance create stronger incentives to limit n?

[17] Rémy Fontenay, Paula Gobbi, and Marc Sangnier (2025), "Land inheritance customs and fertility," working paper. Thierry Gay, Paula Gobbi, and Marc Sangnier (2025), "Inheritance rules and fertility: evidence from French regions," working paper.

16 Social norms and outsourced childcare

Motivation. Market childcare (daycare, nannies) is often stigmatized or judged relative to maternal care. Social norms can therefore slow the adoption of outsourced care even when it is privately efficient.

Mechanism. Households choose the share s of childcare purchased on the market. Full income rises when $w_f > p_s$ (buying care frees female time cheaply), but deviating from the societal norm s^* generates a quadratic utility loss $(\tau/2)(s - s^*)^2$. Stronger τ pins s closer to s^*; higher female wages tilt optimal s toward market care when $w_f > p_s$. The model is partial equilibrium with $n = 1$ child to keep focus on the intensive margin of outsourcing.

References. Doepke *et al.* (2023, Sec. 5.3); Hazan, Weiss, and Zoabi (2021).[18]

Variables and parameters. $s \in [0, 1]$: share of childcare purchased; p_s: price of outsourced childcare; ϕ: total childcare time per child; w_m, w_f: wages; $\tau > 0$: strength of the norm; $s^* \in [0, 1]$: reference level of s. (Fix $n = 1$ child.)

Objective. With one child, indirect utility as a function of s is

$$\max_{s\in[0,1]} \quad \underbrace{w_m + w_f\big(1 - (1 - s)\phi\big) - sp_s\phi}_{\text{full income minus childcare spending}} - \frac{\tau}{2}(s - s^*)^2.$$

Constraints.

- Choice set: $s \in [0, 1]$.
- Parameters: $\tau > 0$, $p_s, \phi, w_m, w_f > 0$, $s^* \in [0, 1]$.

1. Show that an unconstrained interior critical point satisfies $\hat{s} = s^* + \frac{\phi(w_f - p_s)}{\tau}$.

2. For an interior solution, show $\partial\hat{s}/\partial w_f = \phi/\tau > 0$ and $\partial\hat{s}/\partial\tau = -\frac{\phi(w_f - p_s)}{\tau^2}$.

[18] Matthias Doepke *et al.* (2023). Moshe Hazan, David Weiss, and Hosny Zoabi (2021), "An aggregate model of changing sectoral and occupational female employment in the US: 1960–2010," CESifo Working Paper No. 8803.

17 Spousal bargaining and fertility (hard)

Motivation. Fertility is a joint decision. Doepke and Kindermann (2019) argue that in high-income countries the outcome is well described by a *veto* model, but before studying vetoes it is useful to solve the **full-commitment** benchmark in which partners can agree on fertility and on how to split consumption. The algebra below follows the simplified exposition in Doepke *et al.* (2023), Sec. 5.2 (equations (16)–(18) there).[19]

Setup. Two partners $g \in \{f, m\}$ choose whether to have a child $n \in \{0, 1\}$. Preferences are linear:

$$u_g(c_g, n) = c_g + \nu_g \, n,$$

where ν_g is the direct utility from having the child (possibly $\nu_f \neq \nu_m$). Each earns wage w_g. Raising a child requires time $\varphi > 0$; the *opportunity cost* of that time is borne in shares χ_f, χ_m with $\chi_f + \chi_m = 1$ (legislation, biology, or norms). Define the **per-unit time cost** (value of time in childcare):

$$\hat{w} \equiv \chi_f \, w_f + \chi_m \, w_m.$$

Joint consumption enjoys economies of scale $\alpha > 0$: total resources available for consumption and childcare are $(1 + \alpha)(w_m + w_f)$.

Variables and parameters. $c_f, c_m \geq 0$; $n \in \{0, 1\}$; $w_f, w_m > 0$; $\varphi > 0$; $\chi_f, \chi_m \geq 0$ with $\chi_f + \chi_m = 1$; $\nu_f, \nu_m \geq 0$; $\alpha > 0$; $\hat{w} = \chi_f w_f + \chi_m w_m$ as above.

Budget constraint (their equation (16)).

$$c_f + \chi_f \, w_f \, \varphi \, n + c_m + \chi_m \, w_m \, \varphi \, n = (1 + \alpha) \, (w_m + w_f).$$

Equivalently, $c_f + c_m + (\hat{w} \, \varphi) \, n = (1 + \alpha)(w_m + w_f)$.

Outside option (no agreement, no child). If the partners do not cooperate, each consumes only her or his wage:

$$\bar{u}_g(0) = w_g.$$

Bargaining with full commitment. Partners cooperate and split the surplus from joint consumption and from the child. With **equal bargaining weights**, the cooperative outcome is obtained by maximizing the **Nash product** of surplus utilities (geometric average of gains over the outside option): **Objective.**

$$\max_{c_f, c_m, n} \quad \left[u_m(n) - \bar{u}_m(0)\right]^{1/2} \left[u_f(n) - \bar{u}_f(0)\right]^{1/2} \;=\; \left[u_m(n) - w_m\right]^{1/2} \left[u_f(n) - w_f\right]^{1/2}$$

subject to the budget constraint above and $n \in \{0, 1\}$.

Constraints.

- Budget: $c_f + c_m + \hat{w} \, \varphi \, n = (1 + \alpha)(w_m + w_f)$.
- $u_g(n) = c_g + \nu_g \, n$.
- $c_f, c_m \geq 0$, $n \in \{0, 1\}$.

[19] Matthias Doepke, Anne Hannusch, Laura Kindermann, and Michèle Tertilt (2023), "A new era of fertility?" Chap. 4 in *The Economics of Fertility: A New Era*, MIT Press (draft materials). Matthias Doepke and Fabian Kindermann (2019), "Bargaining over babies: theory, evidence, and policy implications," *Journal of the European Economic Association* 17(3): 700–729.

1. (**Equation (17).**) Take n as given (for example $n = 1$). Define the surplus $S_g \equiv u_g(n) - w_g$. Show that the budget implies

$$S_m + S_f = \alpha\,(w_m + w_f) + (\nu_m + \nu_f)\,n - \hat{w}\,\varphi\,n.$$

Hint: Express $S_m + S_f$ using $u_g = c_g + \nu_g n$ and substitute $c_f + c_m$ from the budget.
Hint: For fixed n, the product $S_m^{1/2} S_f^{1/2}$ with $S_m, S_f \geq 0$ and $S_m + S_f$ fixed is maximized when $S_m = S_f$ (AM–GM inequality: the product is largest when the two surpluses are equal).
Deduce that for each $g \in \{f, m\}$,

$$u_g(n) = \underbrace{w_g}_{\text{outside option}} + \underbrace{\frac{\alpha}{2}\,(w_m + w_f)}_{\text{coop surplus}} + \underbrace{\frac{1}{2}(\nu_m + \nu_f - \hat{w}\,\varphi)\,n}_{\text{fertility surplus}}\,. \tag{17}$$

2. (**Equation (18).**) Because n enters both partners' utilities only through the common term $\frac{1}{2}(\nu_m + \nu_f - \hat{w}\varphi)n$ in (17), they agree on fertility if and only if utility is higher at $n = 1$ than at $n = 0$. Show that this is equivalent to

$$\nu_m + \nu_f \geq \hat{w}\,\varphi. \tag{18}$$

Hint: Compare $u_g(1)$ and $u_g(0)$ from (17), or compare total surplus at $n = 1$ and $n = 0$.

Interpretation. Condition (18) says that the **sum** of direct utilities from the child must exceed the **total** opportunity cost of time $\hat{w}\,\varphi$ (not each partner's share separately): cooperation and transfers can compensate an unequal distribution of χ_g, but only if the joint surplus from fertility is nonnegative.

18 Parenting norms and status comparisons

Motivation. In many rich countries, parenting has become more "intensive": tutoring, extracurriculars, and peer competition over children's outcomes. That social pressure can interact with the classic quantity–quality trade-off.

Mechanism. Utility depends on $\ln(q - \chi\tilde{q})$: parents care about quality *relative* to a benchmark $\tilde{q}$ (other families), with strength χ. In symmetric equilibrium $\tilde{q} = q$, status concerns raise the marginal return to s and pull resources toward quality, reducing equilibrium fertility n^{eq} compared to $\chi = 0$. The model is a stylized version of "keeping up" in education spending (Kim *et al.*; Doepke and Zilibotti, 2019).

References. Hannusch and Yum (2026, Sec. 3.2); Kim, Yum, and Yu (2024); Doepke and Zilibotti (2019).[20]

Variables and parameters. $\chi \in [0, 1)$: strength of comparison motives; $\tilde{q} > 0$: benchmark quality (fixed in partial equilibrium).

Objective.

$$\max_{c,\, n,\, q,\, s,\, \ell} \quad \theta_c \ln c + \theta_n \ln n + \theta_q \ln(q - \chi\tilde{q}),$$

with $\theta_c, \theta_n, \theta_q > 0$, $\theta_c + \theta_n + \theta_q = 1$, and $q - \chi\tilde{q} > 0$. **Constraints.**

- $c + sn = (1 - \ell)w, \quad n = \phi\ell, \quad q = \delta s.$
- Assume $\theta_n(1 - \chi) > \theta_q$ for interior solutions.

1. Show that in a symmetric equilibrium with $\tilde{q} = q$, fertility satisfies

$$n^{\text{eq}} = \frac{\phi\big(\theta_n(1 - \chi) - \theta_q\big)}{(1 - \theta_q)(1 - \chi)}.$$

2. Show that $\partial n^{\text{eq}}/\partial\chi < 0$: stronger status concerns lower fertility.

[20] Minchul Kim, Minchul Yum, and Chulhee Yu (2024), "Status seeking and fertility," working paper. Matthias Doepke and Fabrizio Zilibotti (2019), *Love, Money, and Parenting: How Economics Explains the Way We Raise Our Kids*, Princeton University Press.

19 Contraceptive effectiveness and unintended fertility

Motivation. Surveys distinguish *desired* from *realized* family size. In developing countries, a large fraction of pregnancies are reported as unwanted or mistimed; better contraception and family-planning access shrink that gap. Policy analysis therefore needs a link between biological risk, behavioral effort, and realized n.

Mechanism. Let realized births $n = n_d + (1-\kappa)u$: even with desired n_d, a residual unintended component $(1-\kappa)u$ remains when control is imperfect ($\kappa < 1$). Utility depends on **desired** n_d (preferences over intended family size), while the budget uses **realized** n (time and money costs). Raising κ lowers realized n for given n_d but also raises optimal n_d because households need less defensive over-birth to reach their utility target—the comparative statics split across the two margins.

References. Hannusch and Yum (2026, Sec. 4.1); Cavalcanti, Gobbi, and Tertilt (2020).[21]

Variables and parameters. c: consumption; n: *realized* births; $n_d \geq 0$: chosen desired fertility; $Y > 0$: full income; $\gamma > 0$: money–time cost per realized birth; $u > 0$: unintended-fertility shifter; $\kappa \in [0,1]$: contraceptive effectiveness; $\theta \in (0,1)$.

Objective. Parents care about *desired* family size in utility but face a budget in realized births:

$$\max_{c,\, n_d} \quad \theta \ln c + (1-\theta) \ln n_d.$$

Constraints.

- Realized births: $n = n_d + (1-\kappa)\, u$.
- Budget: $c + \gamma\, n = Y$.
- Interiority: $c > 0$, $n_d > 0$, $Y > \gamma n$.

1. Substitute n into the budget and show that the first-order condition implies

$$n_d^* = \frac{(1-\theta)\, Y}{\gamma} - (1-\theta)\,(1-\kappa)\, u, \qquad n^* = \frac{(1-\theta)\, Y}{\gamma} + \theta\,(1-\kappa)\, u.$$

2. Show that $\partial n^*/\partial \kappa = -\theta u < 0$ and $\partial n_d^*/\partial \kappa = (1-\theta)\, u > 0$: better control lowers *realized* total fertility but raises *desired* fertility (households no longer need to "overshoot" to compensate for unintended births).

[21] Tahereh Cavalcanti, Paula Gobbi, and Michèle Tertilt (2020), "Fertility and family planning policies in developing countries," working paper.

20 Religion and doctrinal fertility

Motivation. Empirical work often finds a positive association between religiosity and fertility. Religious traditions differ in explicit pronatalism, but many emphasize continuity, lineage, or a duty to have at least "enough" children. A reduced-form way to encode **doctrine** without utility at $n = 0$ is to measure fertility utility only above a reference level.

Mechanism. Write utility as $\theta \ln c + (1-\theta)\ln(n - \underline{n})$ with $\underline{n} \geq 0$. The argument $n - \underline{n}$ is the number of births *beyond* a doctrinal or cultural floor (e.g. "at least replace yourselves"). The marginal utility of an extra birth is $1/(n-\underline{n})$, which rises as n approaches $\underline{n}$ from above: the first marginal births above the floor are especially valuable spiritually, then diminishing returns set in. The solution shifts fertility upward when $\underline{n}$ rises: higher commanded minima pull n^* up one-for-one at the margin (coefficient θ in $\partial n^*/\partial \underline{n}$).

References. Barro and McCleary (2003) on religiosity and economic outcomes; Iannaccone (1998) on the economics of religion.[22]

Variables and parameters. $c > 0$: consumption; $n > \underline{n}$: total fertility; $\underline{n} \geq 0$: doctrinal floor (parameter); $\theta \in (0,1)$: taste weight on c versus religious–fertility utility; $Y > 0$: full income; $\gamma > 0$: resource cost per child.

Objective.

$$\max_{c,\,n} \quad \theta \ln c + (1-\theta)\ln(n-\underline{n}) \qquad \text{s.t.} \qquad c + \gamma n = Y.$$

Constraints.

- Budget: $c + \gamma n = Y$.
- Interior doctrine: $n > \underline{n}$ and $c > 0$ (equivalently $Y > \gamma \underline{n}$).

1. Show that the interior first-order conditions imply

$$n^* = \frac{(1-\theta)\,Y}{\gamma} + \theta\,\underline{n}, \qquad c^* = \theta\,(Y - \gamma \underline{n}).$$

2. Verify that $n^* > \underline{n}$ if and only if $Y/\gamma > \underline{n}$ (resources suffice to exceed the doctrinal floor).

3. Show that $\partial n^*/\partial \underline{n} = \theta > 0$ and interpret: a higher doctrinal floor raises optimal fertility when the optimum is interior.

[22] Robert J. Barro and Rachel M. McCleary (2003), "Religion and economic growth across countries," *American Sociological Review* 68(5): 760–781. Laurence R. Iannaccone (1998), "Introduction to the economics of religion," *Journal of Economic Literature* 36(3): 1465–1495.